4. User Interface Testing

Check whether labels, buttons, input fields, generated plans and error messages are clear and readable on different screen sizes.

Tabulation

Test Type	What is checked
Functional	Features work as expected
Integration	Frontend/backend/Gemini communication
UI	Layout and readability
Error	Invalid input/API failure handling

Coding / Technical Example

```
def test_plan(plan):
    assert plan is not None
    assert len(plan) > 0
    print("PASS")

test_plan("Sample FitBuddy plan")
```

Picture / Diagram

```
INPUT TEST → FUNCTION TEST → API TEST → UI TEST → FIX → RETEST
```

5. Test Cases

A test case should contain an identifier, input, expected result and actual result/status.

Tabulation

Test Type	What is checked
Functional	Features work as expected
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Coding / Technical Example

```
def test_plan(plan):
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```

Picture / Diagram

```
INPUT TEST → FUNCTION TEST → API TEST → UI TEST → FIX → RETEST
```

6. Error Handling

The application should show a friendly message for missing input, network/API failure or invalid requests instead of exposing technical details.

Tabulation

Test Type	What is checked
Functional	Features work as expected

Test Type	What is checked
Integration	Frontend/backend/Gemini communication
UI	Layout and readability
Error	Invalid input/API failure handling

Coding / Technical Example

```
def test_plan(plan):
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Picture / Diagram

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